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(54) **FLUID TREATMENT TANK HAVING A
LASER WELDED DISTRIBUTOR PLATE**

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8, 2019, now Pat. No. 12,090,573.

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C02F 5/00 (2023.01)

(52) **U.S. Cl.**
CPC **B23K 26/28** (2013.01); **C02F 5/00**
(2013.01); **C02F 2201/002** (2013.01)

(58) **Field of Classification Search**

None

See application file for complete search history.

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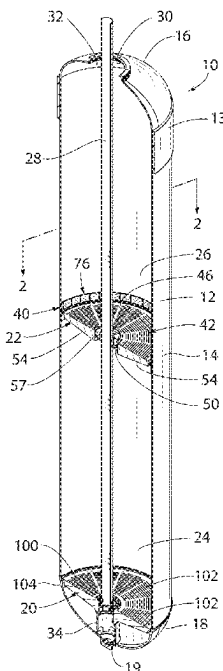
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(57) **ABSTRACT**

A distributor plate for a fluid treatment tank such as a water softener pressure vessel separates the resin bed from a lower end of the resin tank. The outer peripheral surface of the distributor plate is laser-welded to an inner peripheral surface of the tank. The outer peripheral surface of the distributor plate may have a biasing mechanism that biases the outer peripheral surface of the distributor plate against the inner peripheral surface of the tank. The biasing mechanism may include a plurality of peripherally-spaced resilient fingers extending axially and radially away from a circular base of an outer ring of the distributor plate into engagement with the inner peripheral surface of the tank liner, each of the fingers having a base connected to the base of the ring and having a free end. Also disclosed is a method of making a fluid treatment tank.

13 Claims, 5 Drawing Sheets



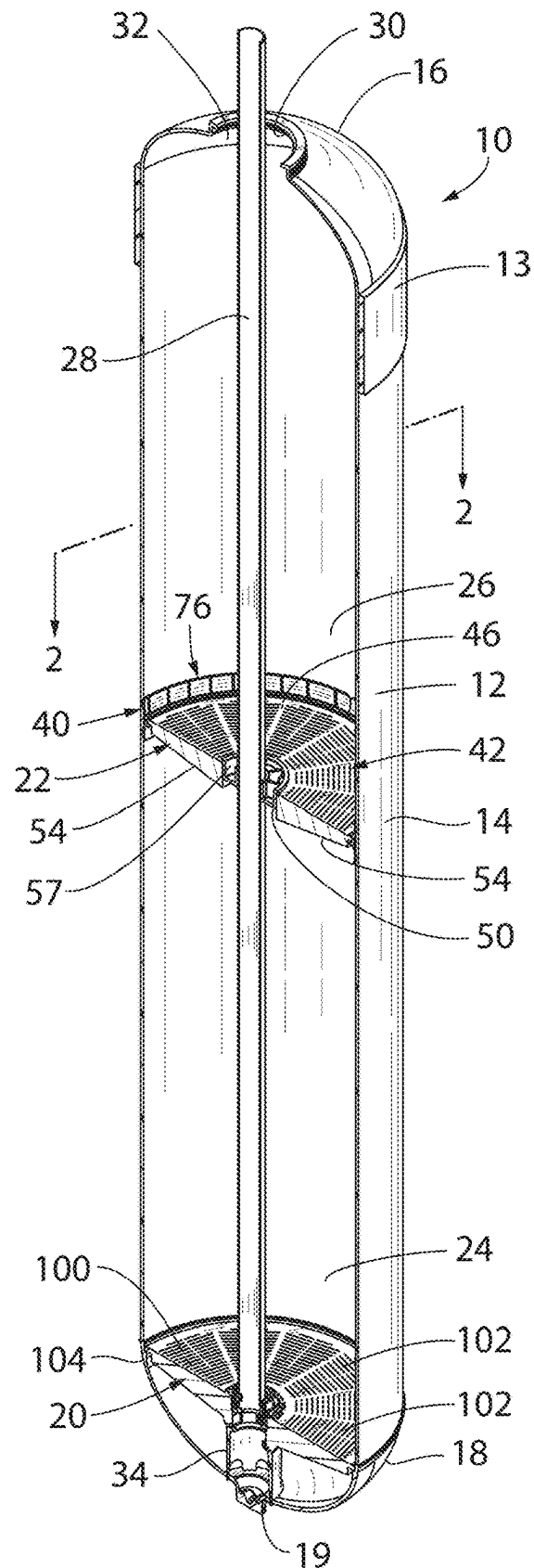


FIG. 1

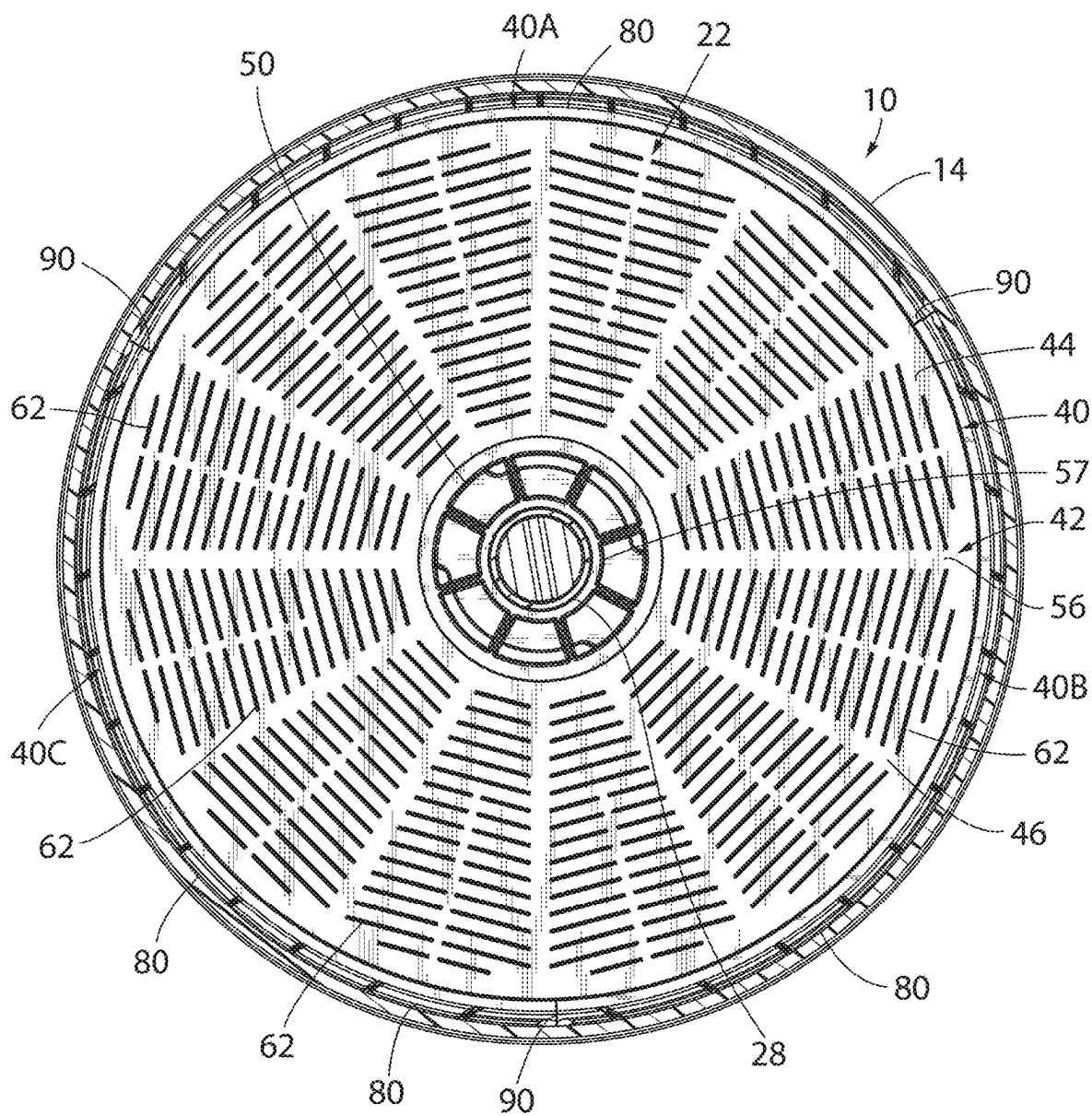


FIG. 2

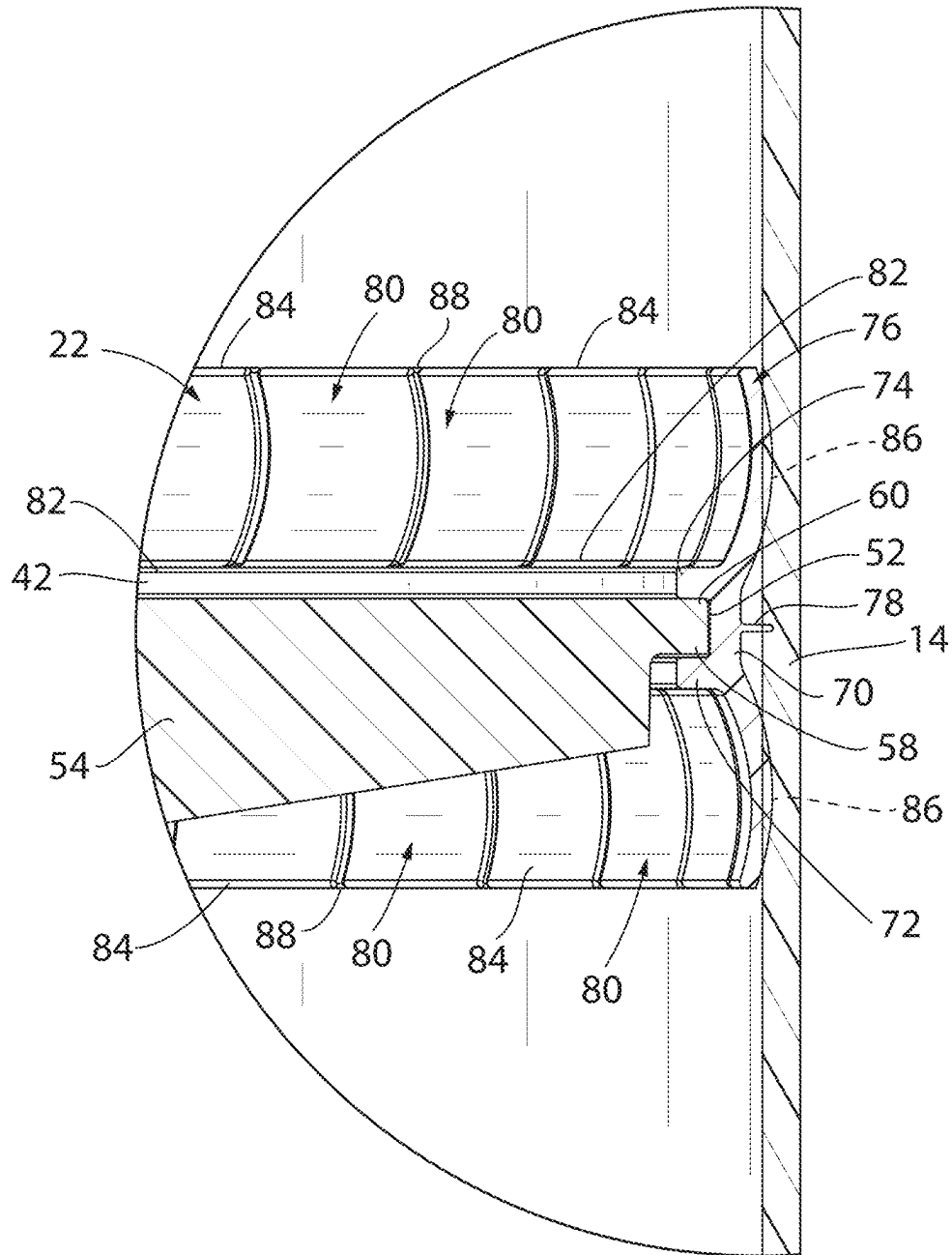


FIG. 3

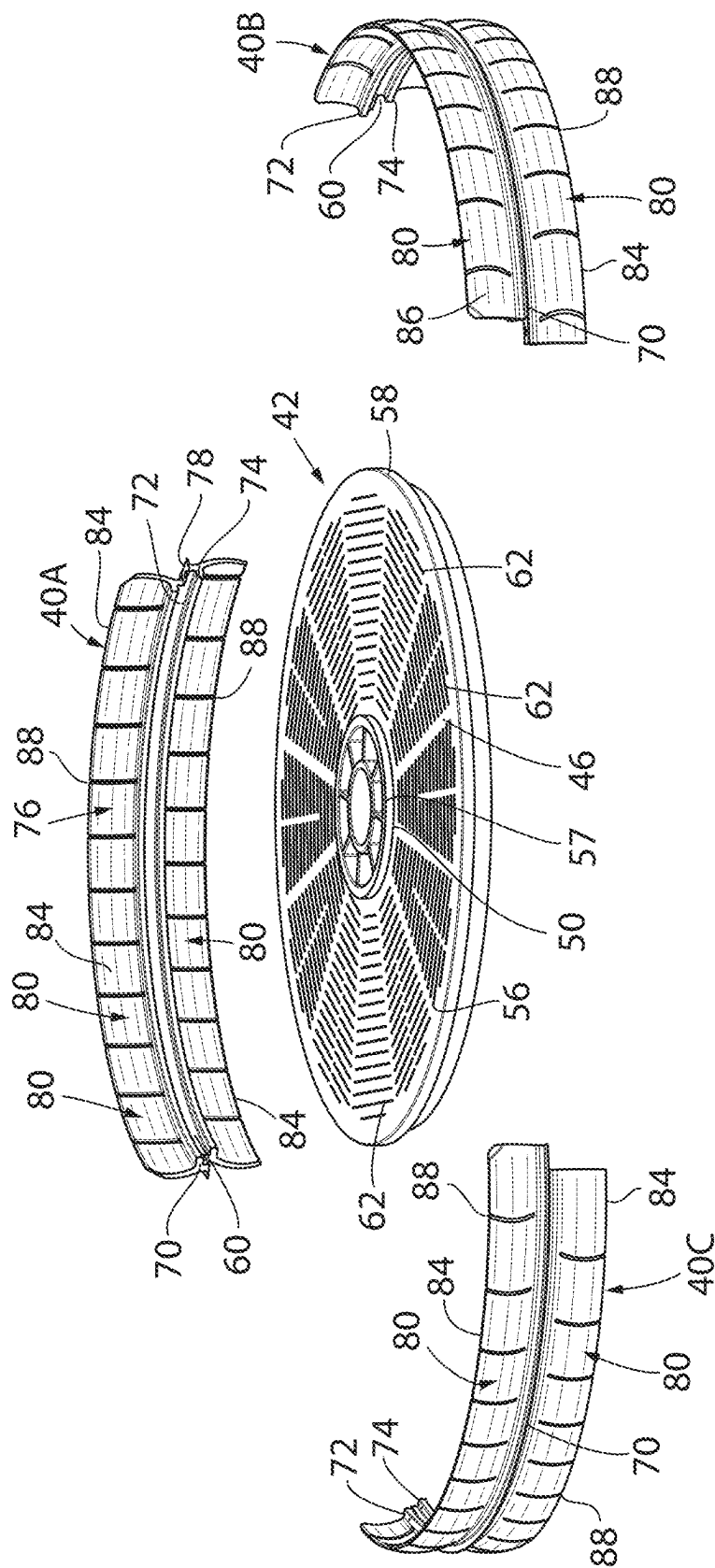


FIG. 4

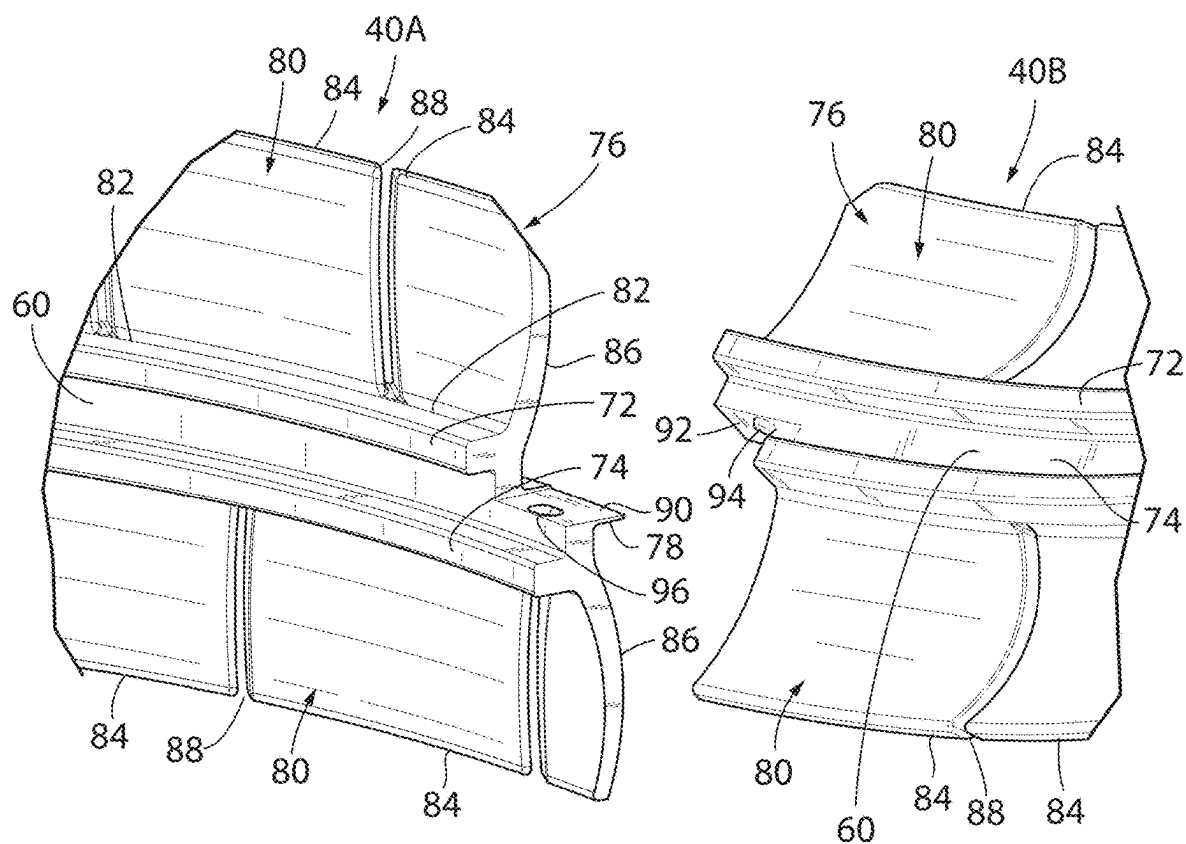


FIG. 5

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FLUID TREATMENT TANK HAVING A LASER WELDED DISTRIBUTOR PLATE

CROSS REFERENCE TO A RELATED APPLICATION

This application is a divisional of U.S. patent application Ser. No. 16/406,576, filed May 8, 2019 and entitled Fluid Treatment Tank Having A Laser Welded Distributor Plate, the contents of which are hereby incorporated by reference in their entirety.

BACKGROUND OF THE INVENTION

1. Field of the Invention

The invention generally relates to fluid treatment tanks and, more particularly, to a tank that has a fluid-permeable distributor plate and that can be easily manufactured and assembled. The invention additionally relates to such a distributor plate and to a method of making a fluid treatment tank including such a distributor plate.

2. Discussion of the Related Art

Fluid-permeable plates, generally known as “distributor plates,” are widely used in a variety of fluid treatment tanks. Typical of these tanks are known as “pressure vessels” in which the fluid being treated is pressurized, albeit usually at a low pressure such as residential water line pressure. One such tank is a pressure vessel known as a “resin tank” of a water treatment system. The typical resin tank is a hollow cylindrical structure the interior of which defines a resin bed configured to hold “resin” and water or another liquid therein. The resin may be provided in the form of a plurality of plastic, e.g., polystyrene, beads. The resin bed is separated from the bottom of the tank by a slotted, perforated, or otherwise fluid permeable distributor plate that permits bidirectional fluid flow therethrough, but which prevents resin from falling through the distributor plate. A riser tube may be centrally positioned within the tank. The typical riser tube extends from the distributor plate to an upper opening in the tank through which treated liquid exits the resin tank. The tank may include a blow-molded plastic tank liner reinforced by an outer layer of fiberglass wrap.

Some pressure vessels have distributor plates mounted well-above the bottom of the tank so as to permit a single tank to contain two or more diverse treatment media. Some of upper plates are located in the vicinity of the vertical center of the tank and thus as known as “mid-plate distributor plates” or simply “mid-plates.” Some tanks contain two more such plates located well-above the bottom of the tank in a vertically-spaced relationship. For the sake of conciseness, all such plates are referred to herein as “mid-plates”, it being understood that the plate need not be located at or near the center of the tank. These mid-plates usually include a cylindrical outer surface that is spin-welded to the interior of the tank’s plastic liner. Such plates are described, for example, in U.S. Pat. Nos. 7,901,576 and 8,382,994 to Enpress.

Spin welding involves imparting relative rotation to the components to be welded to generate sufficient frictional heat to the parts at their interface to melt the mating surface of one or both components. Once rotation stops, the material cools and solidifies and fuses the components together, forming a welded assembly. Spin welding necessarily involves holding the two components in close concentric

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engagement while they are rotated relative to one another. In the case of a water treatment tank mid-plate, spin-welding typically uses a rotating mandrel or spindle that engages the mid plate while the tank liner is held stationary. Spin welding a distributor plate to the interior of a tank liner is difficult or impossible unless the tank liner is open near the plate so as to allow insertion of the mandrel axially into the tank. The Enpress patents discussed herein propose first forming the tank liner, then cutting the tank liner apart in sections to provide an open end near the mid plate, then spin-welding the mid plate to the inner surface of the tank liner, then rejoining the tank liner sections, presumably by thermally bonding them together. The resultant process is time-consuming and equipment intensive. It also risks loss of tank integrity where the cut tank liner sections are rejoined if the welds are imperfect. It also risks misalignment of the rejoined sections unless care is taken to maintain concentricity of those sections during the rejoining process, a condition that can be difficult to meet given the tendency of the liner to become out of round during handling.

The inventor has realized that the need to section tank liners theoretically could be avoided by laser welding the tank liner and distributor plate together. Laser welding involves directing a laser beam through a first material that is relatively transparent to the beam to a second material, that is relatively absorbent, so as to heat the second material sufficiently to melt and fuse it to first material upon subsequent cooling. Since the weld energy is imparted through the transparent outer component, that component need not be open in the vicinity of the inner component to accommodate a rotating mandrel or spindle. However, laser welding is similar to spin welding to the extent that requires that the components being welded be held in close engagement with one another during the laser welding process. The inventor has recognized that, in the case of treatment tank mid plate or other distributor plate, incorporating any measures into a welding system that clamps the plate to the tank liner or otherwise holds the plate and the tank liner in engagement with one another would have operate from an open end of the tank liner—just as with a spin welder—negating a major potential benefit of laser welding over spin welding.

The need therefore has arisen to provide a mechanism and/or process for holding an outer peripheral surface of a mid-plate or other distributor plate closely adjacent an inner peripheral surface of a water treatment tank preparatory to and during a laser welding process.

The need additionally has arisen to provide a mechanism and/or a process for welding a distributor plate in a water treatment tank without having to leave an end of the tank adjacent the distributor plate open during the welding process.

The need additional has arisen to provide a water treatment tank with welded distributor plate and a seamless tank in the vicinity of the distributor plate.

SUMMARY

In accordance with a first aspect of the invention, one or more of the above-identified needs is met by providing a fluid treatment tank comprising a tank at least one distributor plate that is configured to support a bed of a fluid treatment media. The tank has an at least generally cylindrical inner surface, and at least one distributor plate received in the tank and configured to separate the bed from a lower end of the tank. The distributor plate has an upper surface, a lower surface, and an outer peripheral surface. The upper surface forms a support surface for fluid treatment media. A plurality

of fluid permeable openings are formed through the distributor plate from the upper surface to the lower surface for the passage of a liquid being treated by the treatment media. The outer peripheral surface of the distributor plate is laser-welded to the inner peripheral surface of the tank.

In order to facilitate welding of the distributor plate to the tank, the outer peripheral surface of the distributor plate may have a biasing mechanism that biases the outer peripheral surface of the distributor plate against the inner peripheral surface of the tank.

The distributor plate may be formed of two or more components including an inner fluid permeable disk and an outer ring that supports the disk in the tank. The biasing mechanism may include a plurality of peripherally-spaced resilient fingers extending axially and radially away from a circular base of the support ring into engagement with the inner peripheral surface of the tank liner, each of the fingers having a base connected to the base of the ring and having a free end. The fingers may be spaced from one another at the free ends thereof, in which case a peripheral seal may extend outwardly from the base of the ring and seal against the inner peripheral surface of the tank beneath the free ends of the fingers. Each of the fingers may be curved along at least a majority of a length thereof and have an apex between the base and the free end thereof, the apex engaging the inner peripheral surface of the tank.

The tank may be made of a first material that is relatively transparent to laser light of a designated frequency, and at least an outer peripheral portion of the distributor plate may be made of a second material that is relatively absorbent to laser light of the designated frequency.

In accordance with another aspect of the invention, a method of forming a fluid treatment tank that has a distributor plate constructed at least generally as discussed above is provided. The method includes engaging an outer surface of the distributor plate with an inner peripheral surface of a fluid treatment tank, the distributor plate being configured to separate a bed of treatment media from a lower end of the tank. The distributor plate has an upper surface, a lower surface, and an outer peripheral surface, the upper surface forming a support surface for the fluid treatment media, and a plurality of fluid permeable openings are formed through the distributor plate from the upper surface to the lower surface for the passage of a liquid being treated by the treatment media. The method additionally includes laser-welding the outer surface of the distributor plate to the inner wall of the tank.

The engaging may comprise deflecting a biasing mechanism on the distributor plate against the inner peripheral wall of the tank. In this eventuality, the engaging may comprise deflecting a circumferentially-spaced resilient fingers against the inner peripheral wall of the tank, the fingers extending radially and axially away from a base of an outer ring of the distributor plate. The distributor plate may have two or more components including an inner disk and an outer support ring of different materials and that includes the fingers.

Various other features, embodiments and alternatives of the present invention will be made apparent from the following detailed description taken together with the drawings. It should be understood, however, that the detailed description and specific examples, while indicating preferred embodiments of the invention, are given by way of illustration and not limitation. Many changes and modifications could be made within the scope of the present invention without departing from the spirit thereof, and the invention includes all such modifications.

BRIEF DESCRIPTION OF THE DRAWINGS

Preferred exemplary embodiments of the invention are illustrated in the accompanying drawings, in which like reference numerals represent like parts throughout, and in which:

FIG. 1 is a sectional side elevation view of a fluid treatment tank including a distributor plate according to an embodiment of the present invention;

FIG. 2 is a sectional end elevation view of the fluid treatment tank of FIG. 1, taken generally along the line 2-2 in FIG. 1;

FIG. 3 is a fragmentary isometric view side view of a portion of the pressure vessel, showing the connection of the distributor plate with the inner peripheral wall of a tank liner of the fluid treatment tank;

FIG. 4 an exploded isometric view of the distributor plate of FIGS. 1-3; and;

FIG. 5 is an exploded fragmentary perspective view of the mating ends of two ring segments of the distributor plate of FIGS. 1-4.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

Distributor plates constructed in accordance with the invention could be used in a variety of fluid treatment tanks configured to treat a variety of fluids using any of a number of media. Hence, while preferred embodiments of the invention now will be described in conjunction with a resin tank of a water treatment system, it is to be understood that the invention is not so limited.

Referring now to the drawings and, initially, FIG. 1, a resin tank 10 includes a tank liner 12 reinforced with wound fiberglass 13 (only a portion of which is shown in FIG. 1). One of variety of water treatment systems with which the resin tank 10 is usable is disclosed in U.S. Pat. No. 6,402,944, the contents which are incorporated herein by reference. It should be noted that, in some applications, the tank need not have an inner liner. The term "tank" therefore should be understood to encompass tanks both with and without an inner tank liner.

The tank liner 12 is made of a thermoplastic material such as blow-molded high-density polyethylene (HDPE). Liner 12 includes a generally cylindrical hollow body or wall 14 and upper and lower generally semi-spherical top and bottom ends 16 and 18. A feature 19 is integrally molded to the bottom end 18 of the liner 12 to rotationally couple the liner 12 to a filament winding machine during the fiberglass winding process. The body 14 may have an internal diameter of, for example, 0.2 m to 1.0 m and a height of 0.5 m to 2 m. The illustrated example has an internal diameter of about 0.25 m. At least one distributor plate is provided in the liner 12 to define a treatment media area thereabove. The illustrated embodiment includes two such plates 20, 22 located near the bottom of the tank 10 and near the center of the tank 10, respectively and defining respective treatment media storage areas 24, 26, thereabove. Each distributor plate 20, 22 has fluid permeable slots or other openings formed through them that are sized and shaped to permit the relatively free flow of the liquid being treated while inhibiting or preventing the ingress of resin beads or other treatment media particles. Each area 24, 26 is configured to receive a bed or layer of a resin or other water treatment media during use of the tank 10. The resin may be provided in the form of a plurality of plastic, e.g., polystyrene, beads. The same or different treatment media may be received in

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each area **24**, **26**. The plate **20** is located near the bottom of the tank **10** and can be thought of as a “bottom plate” distributor. The plate **22** is located above the bottom dome **18** and thus can be thought of as a “mid-plate” distributor plate. A riser tube **28** extends vertically through the center of the tank **10** from the beneath the bottom plate **20**, extends through center openings in both plates **20** and **22**, and extends to or through an upper center upper opening **30** of the tank **10**. An annular opening **32** is formed between the outer peripheral surface of the riser tube **28** and the perimeter of the opening **30**. An inlet tube **34**, positioned beneath the lower end of the riser tube **28**, extends downwardly through the bottom distributor plate **20** in communication with the lower end portion of the tank **10**. During a water treatment operation, untreated water enters the tank **10** from above through the annular opening **32**, flows through the media areas **26** and **24** and the openings in the associated distributor plates **22** and **20** in succession, enters the inlet tube **34**, and rises out of the tank **10** through the riser tube **28**. This flow may be reversed, for example, during a treatment media generation cycle.

Pursuant to an implementation of the invention, one or both of the distributor plates **20** and **22** is mounted in the tank liner **12** by laser welding. In the illustrated embodiment, the mid-plate **22** is mounted in the tank liner **12** by laser welding, it being understood that, in other implementation/s, the bottom plate **20** and/or still other plates could be mounted in the tank liner by laser welding instead of or in addition to the mid-plate **20**.

With additional reference now to FIGS. **1** and **2**, the mid-plate **22** is shown in additional detail. The plate **22** is a two-component assembly including an outer ring **40** mounted on the tank liner wall **14** as described below and an inner disk **42** supported on the ring **40**. This two-part distributor plate construction 1) permits the ring **40** to be injection-molded from a thermoplastic material that is relatively-easily thermally-bonded to the tank liner **12** but that has a relatively low dimensional predictability, while 2) permitting the disk **42** to be injection-molded from a thermoplastic material that cannot be easily welded to the material of the tank liner **12** but that has a relatively high dimensional predictability. “Dimensional predictability” in this regard refers to a material that can be injection-molded with fine features of on the order of a less than a centimeter and as fine as a few tenths of a millimeter with close tolerances and with high repeatability on a shot-by-shot basis over a relatively large number of shots. High “dimensional predictability” is generally inversely related to shrink rate and is even more closely inversely related to shrink rate variability. Forming the disk **42** from a material having a relatively high dimensional predictability when compared to that of a material that can thermally bonded to the tank liner **12** permits the slots and other fine features on the distributor plate **20** to be made reliably with close tolerances.

The inner disk **42** of the mid-plate **22** of this embodiment is made from unfilled or glass-filled Noryl® (Noryl® is a registered trademark of Sabic Innovative Plastics IP B.V.), which is family of amorphous reinforced materials in the form of modified polypropylene ether (PPO) resins that are formed from amorphous blends of PPO resin and polystyrene. The characteristics of Noryl and other possible inner disk materials, as well as the design and construction of such disks, are described in great detail in U.S. Pat. No. 10,106,434, the subject matter of which is hereby incorporated by reference in its entirety.

Referring particularly to FIGS. **1-3**, the disk **42** includes upper an upper fluid media support surface **46**, a lower

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surface, a central hub portion **50** through which the riser tube **28** passes, a cylindrical outer radial edge **52**, a plurality of circumferentially-spaced reinforcing ribs **54**, and a slotted plate portion **56**. An annular plug **57** is received in the hub portion **50**. The plug **57** engages the periphery of the hub portion at its outer perimeter and the riser tube **28** at its inner perimeter. The slotted plate portion **56** is supported on the ribs **54** and extends from the hub portion **50** to the outer ring radial end **52**. The ribs **54** extend at least generally radially from the hub portion **50** to the vicinity of the outer radial edge **52** but stop short of the edge **52** of the disk **42** to form an annular tongue **58** received in an inner groove **60** in the outer ring **40**.

Referring now to FIGS. **2-5**, the above-described openings are provided in the form of a plurality of slots **62** that are formed in the slotted plate portion **56** of distributor plate **22**. The slots **62** are sized, shaped, and orientated to promote uniform fluid flow through the distributor plate **22** and the resin without being clogged with the resin beads or other media particles. The slots **62** could be relatively straight slots that extend either radially or tangentially, or could be curvilinear, such as generally spiral-shaped. The “slot density” or proportion of the total area of all slots **62** to the total area of the slotted plate portion **56** also can vary considerably from application to application. In addition, one or more of the slot size, slot shape, slot orientation, and slot density can vary across the radius of a given slotted plate portion **56** to contribute to a desired flow distribution across the radius of the tank **10**. Each slot **62** ideally should be wide enough to permit the fluid being treated and other fluids such as brine to pass therethrough but narrower than the smallest diameter bead or other particle forming the treatment media, it being understood that the diameters of such particles vary greatly and that it is difficult, if not impossible, to design slots or other openings that prevent any particles from falling through.

Still referring to FIGS. **3** and **4**, the tank liner **12** and outer ring **40**, or at least those portions of the outer ring **40** that are welded to the tank liner **12**, are made of materials that are suitable for the laser-welding of the ring **40** to the tank liner **12**. In order to make the tank liner **12** and ring **40** from as similar of materials as practical, the tank liner **12** may be made from a material of a molecular weight that is on the high end of the range of suitable molecular weights for blow-molding, and the material of the ring **40** is of a molecular weight that is on the low end of the range of suitable molecular weights for blow-molding that can be injection-molded.

Suitable ring materials are those that are fusible to the tank liner **12** and that are relatively absorbent to laser energy of a wavelength found in laser welders. In the illustrated embodiment in which the tank liner **12** is made of a blow-molded, high-density polyethylene (HDPE) that is relatively transparent to laser energy. One such material is available from Exxon Mobile under the PAXON® line and, more specifically PAXON BA53-055. The ring **40** also is preferably made from a high-density polyethylene (HDPE). A suitable material is available from LyondellBasell under the Petrothene® line and, more specifically, Petrothene LT570401. The ring material is impregnated with a material, such as carbon black, that renders it absorbent to laser energy. The laser energy absorbent material may be mixed with the HDPE resin prior to the injection molding process in a percentage sufficient to achieve the desired degree of melting during the laser welding process.

Referring briefly to FIGS. **4** and **5**, the outer ring **40** is formed of a plurality (three in this embodiment) arcuate ring

segments **40A**, **40B**, and **40C** connected end to end as described below. Each segment could be formed of two or more interconnected pieces but, in the illustrated embodiment, is formed as a single injection molded component. It thus should be understood that the discussion of the “ring” that follows refer to the fully-assembled ring **40** formed by interconnecting the ring segments **40A**, **40B**, and **40C**.

Referring to FIGS. 2-5, the ring **40** includes a circular base **70** having an outer peripheral surface, annular horizontal flanges **72** and **74** extending radially inwardly from the cylindrical base **70** to form the channel **60** therebetween, and a biasing mechanism or spring mechanism **76** for holding the distributor plate **22** in the tank liner **12** before and during the laser welding operation, and an annular seal **78**. The spring mechanism **76** could be formed integrally with or separately from the base **70**. It could comprise a single bellows or similar structure extending completely around the mid-plate **22** or two or more spring elements spaced circumferentially of one another. In the illustrated embodiment, the spring mechanism **76** comprises fingers **80** collectively extending around the entire circumference of the distributor plate **22**. Two (upper and lower) sets of arcuate fingers **80** are provided in this embodiment, with the sets being symmetrical about a radial bisector of the base **70**. Each set contains 30-40 fingers.

Referring especially to FIGS. 3-5, the fingers **80** of this embodiment are molded integrally with the remainder of the ring **40**. Each finger **80** has a base **82** merging into the base **70** of the ring **40** and a free end **84** located axially and radially apart from the base **82**. Each finger **80** is concave, presenting an apex **86** located between the base **82** and the free end **84**. Each finger **80** engages the inner peripheral surface of the tank liner **12** at least at its apex **86**, causing the finger **80** to resiliently deflect at its base **80** upon engagement of the with the inner wall of the tank liner **12** and impose a retention force on the tank liner **12**. Adjacent fingers **80** could be coupled to one another by reduced-thickness portions of ring material but, in the illustrated embodiment, are spaced by a small gap **88** to enhance the resiliency of the individual fingers **80** and to permit one finger **80** to deflect more than an adjacent finger **80**, better-accommodating out of roundness, ridges, or other surface irregularities in the inner peripheral surface of the tank liner **12**. In the present embodiment, each finger **80** is 17.5 mm high, measured axially, from base to tip, 28 mm wide, and 1.5 mm thick. Each gap **88** between adjacent fingers **80** is 0.78 mm thick. In the free state of the fingers, the maximum diameter of the ring **40**, as measured at the apices **86** of the fingers **80**, is greater than the inner diameter of the tank liner body **14** by at least 0.1 cm, more typically on the order of at least 0.3 cm, and possibly on the order of 0.5 cm or more.

Still referring to FIG. 3, the peripheral seal **78** extends outwardly from the base **70** of the ring **40** to seal the ring seals against the inner peripheral surface of the tank liner body **14**, preventing fluid and treatment media from flowing through the gaps **88** between the fingers **80** and bypassing the distributor plate **22**. The illustrated seal **78** comprises a relatively thin annular flange extending radially from the radial bisector of the ring **40** into engagement with the inner peripheral surface of the tank liner **12**. This seal **78** could be formed integrally with the ring **40** as illustrated or could be mounted in the outer peripheral surface of the ring **40**.

As mentioned above, the ring **40** is constructed in segments **40A**, **40B**, **40C** so as to permit it to be formed around the disk **42** so as to capture the tongue **58** of the disk **42** in the groove **60** of the ring **40**. Each segment **40A**, **40B**, **40C** extends about 120 degrees around the plate **22**. Adjacent

segments could be connected to one another in any desired manner. As seen in FIGS. 2 and 5, adjacent segments of this embodiment have stepped ends that mate with one another at a stepped seam **90**. At the mating ends, one of the segments has a first, upwardly facing flat surface **92**, and the mating segment has a second, downwardly facing flat surface **94**. A lug or projection **96** on one of the surfaces is received in a pocket or recess **98** in the other surface when the adjacent segments are connected to hold the segments together.

To mount the mid-plate **22** in the tank liner **12**, the mid-plate **22** is pre-assembled by connecting the ring segments **40A**, **40B**, **40C** to one another with the tongue **58** on the disk **42** captured in the groove **70** in the ring **40**. The plate is then inserted into the cylindrical tank liner body **14** from one of the ends to the desired position. During this process, the fingers **80** deflect to permit axial movement of the mid-plate **22** relative to the tank liner body **14**, albeit with some resistance. When the mid-plate **22** is properly positioned in the tank liner body **14**, the fingers **80** hold the mid-plate **22** in place, with at least the apices **86** of the fingers **80** firmly engaging the inner peripheral surface of the tank liner body **14**. As mentioned above, finger deflection also can accommodate signifying out-out roundness and/or other surface irregularities formed in the inner peripheral surface of the tank liner body **14**.

The ring **40** then is laser welded to the tank liner body **14** using a laser welder that transmits laser energy through the tank liner body **14** from the outside. A laser suitable for this process is commercially available from Leister under the NOVALSTM Basic AT name. This laser has a 200W laser line and operates at a wavelength of 975 nm. The laser energy passes through the tank liner body **14** without appreciably heating it and is absorbed in the ring **40**, heating at least the outer portions of fingers **80** to above their melting point. When the laser is turned off, the melted material cools, fusing the fingers **80** to the inner peripheral surface of the tank liner body **14** in the vicinity of the apices **86**, and possibly elsewhere. The mid-plate **22** now is retained in place with sufficient force to support a bed of treatment media and to withstand the pressures encountered during typical use of the tank **10**.

Because laser welding does not require the insertion of a rotating mandrel or comparable structure into the tank liner body **14**, the tank liner body **14** need not be cut in the vicinity of the mounting location prior to the welding and rejoined after the welding operation. The tank liner body **14** instead can be seamless between the end domes **16** and **18**. Fabrication therefore is substantially simplified when compared to the fabrication of a tank having a mid-plate installed by spin-welding. Tank dimensional reliability and tank integrity also are enhanced due to the lack of any seam along the cylindrical portion of the tank liner body **14**.

Referring again to FIGS. 1 and 2, the bottom-plate **20** also has a disk **100** formed Noryl or another material having high dimensional predictability, and has slots **102** formed there-through that may have the same or similar characteristics of the mid-plate slots **62**. The bottom-plate **20** also could have a ring that is laser-welded to the inner periphery of the tank liner, likely at a location just above the bottom dome **18**, in the same manner as the mid-plate **22**. Alternately, the bottom plate **20** could have a ring **18** that has ends that are fusion bonded to facing ends of the tank liner body **14** and to the bottom dome **18** as described in U.S. Pat. No. 10,106,434, described above and incorporated by reference in its entirety. However, in the illustrated embodiment, an outer ring is omitted in its entirety. The outer peripheral edge of the disk

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100 instead rests on a support and is prevented from significant upward movement by a weld bead formed by welding the upper end of the bottom dome 18 to the lower end of the tank liner body 14. That weld bead projects over the upper surface of the disk typically about 0.5 mm to 1.0 mm, preventing significant disk upward surface. The support could be formed from a pan or other structure in the dome 18 and/or, as is the case in the illustrated embodiment, by a step or shelf 104 provided in or on the upper end portion of the dome 18.

Although the best mode contemplated by the inventors of carrying out the present invention is disclosed above, practice of the present invention is not limited thereto. It will be manifest that various additions, modifications and rearrangements of the aspects and features of the present invention may be made in addition to those described above without deviating from the spirit and scope of the underlying inventive concept.

For example, distributor plates and tanks as described herein could be used to store fluid treatment media other than resin and could be used to treat fluids other than water.

In addition, it is conceivable that the entire distributor plate, including the fingers or other biasing mechanism, could be formed of a single component formed from a material that can be laser welded to the tank liner, in which case the disk and the attendant openings might not have the same degree of dimensional stability as discussed above.

The scope of some changes to the described embodiments is discussed above. The scope of other changes to the described embodiments that fall within the present invention but that are not specifically discussed above will become apparent from the appended claims.

We claim:

1. A method comprising:

engaging an outer peripheral surface of a distributor plate with an inner peripheral surface of a fluid treatment tank, the distributor plate being configured to separate a bed of treatment media from a lower end of the tank, the distributor plate having an upper surface, a lower surface, and an outer peripheral surface, and a plurality of fluid permeable openings formed through the distributor plate from the upper surface to the lower surface for the passage of a liquid being treated by the treatment media, wherein the tank is made of a first material and at least an outer peripheral portion of the distributor plate is made of a second material that is more absorbent to laser light of a designated frequency than the first material; and

laser-welding the outer peripheral surface of the distributor plate to the inner peripheral surface of the tank using a laser welder that transmits laser energy through the tank from outside of an outer peripheral surface of the tank, wherein

the engaging comprises deflecting a biasing mechanism on the distributor plate against the inner peripheral surface of the tank.

2. The method of claim 1, wherein the engaging comprises deflecting circumferentially-spaced resilient fingers against the inner peripheral surface of the tank, the fingers extending radially and axially away from a base of an outer ring of the distributor plate.

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3. The method of claim 2, wherein adjacent fingers are spaced from each other at least at free ends thereof, and further comprising sealing the base of the ring against the inner peripheral surface of the tank at a location radially inboard from the free ends of the fingers.

4. The method of claim 1, wherein the tank and the outer peripheral surface of the distributor plates are made from different high-density polyethylene materials.

5. The method of claim 1, wherein the material forming the outer peripheral surface of the distributor plate is impregnated with carbon black.

6. The method of claim 1, wherein the tank includes a tubular body that is unsegmented between upper and lower ends thereof, and wherein the distributor plate is a mid-plate that is located above an end plate, the end plate being located near the lower end of the tubular body.

7. The method of claim 6, further comprising, after the laser welding, attaching domes to the upper and lower ends of the tubular body.

8. The method of claim 1, further comprising, after the laser welding, overwrapping the outer peripheral surface of the tank with fiberglass.

9. The method of claim 1, wherein the distributor plate comprises an outer ring and an inner disk that is supported on the ring and through which the openings are formed, the disk being formed of a different material than the ring.

10. A method comprising:

providing an unsegmented tubular tank liner body having open upper and lower ends, the body being formed from a first high-density polyethylene material,

positioning a mid-plate distributor between the upper and lower ends of the body and biasing an outer peripheral surface of the mid plate distributor into engagement with an inner peripheral surface of the body, the outer peripheral surface of the mid-plate distributor being formed from a second high-density polyethylene material that is less absorbent to laser light of a designated frequency than the first high-density polyethylene material; then

laser-welding the outer peripheral surface of the mid-plate distributor to the inner peripheral surface of the body using a laser welder that transmits laser energy through the body from outside of an outer peripheral surface of the body;

attaching an end-plate distributor to the body beneath the mid-plate distributor;

attaching domes to the upper and lower ends of the body to form a tank liner; and

overwrapping the outer peripheral surface of the body with fiberglass.

11. The method of claim 10, wherein the biasing comprises deflecting circumferentially-spaced resilient fingers against the inner peripheral surface of the body.

12. The method of claim 10, wherein the outer peripheral surface of the mid-plate distributor is impregnated with carbon black.

13. The method of claim 10, wherein the mid-plate distributor comprises an outer ring and an inner disk that is supported on the ring and through which the openings are formed, the disk being formed of a different material than the ring.

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